



RAFFLES INSTITUTION
H2 Computing (9569)
2025 Year 5

Tutorial S1A: Algorithmic Representation

Section A (Discussion Questions)

1. **[flowchart]** Draw a flowchart for solving the quadratic equation $Ax^2 + Bx + C = 0$ where A does not equal 0.
2. **[flowchart]** Draw a flowchart that counts the number of digits in an integer.
3. **[flowchart]** Draw a flowchart to sum the digits in an integer.
4. **[flowchart]** A man is paid 1¢ for his job on the first day, 2¢ for the second day, 4¢ the third day, and so on, doubling each day on the job for 30 days. Draw a flowchart to calculate his wages on the 30th day and his total for 30 days.
5. **[flowchart, variables]** A marathon runner records their time for a race in hours, minutes and seconds. An algorithm is shown below in structured English.

INPUT race time as hours, minutes and seconds
CALCULATE race time in seconds
STORE race time in seconds
OUTPUT race time in seconds

- (a) State and describe the variables required for this algorithm.

Identifier	Description
RaceHours	The hours part of the race time.

- (b) Before the program is written, the design is amended. The new design includes input of the runner's current personal best marathon time (in seconds).

The output will now also show one of the following messages:

- "Personal best time is unchanged"
- "New personal best time"
- "Equals personal best time"

- (i) State and describe the additional variable needed for the new design.

Identifier	Description

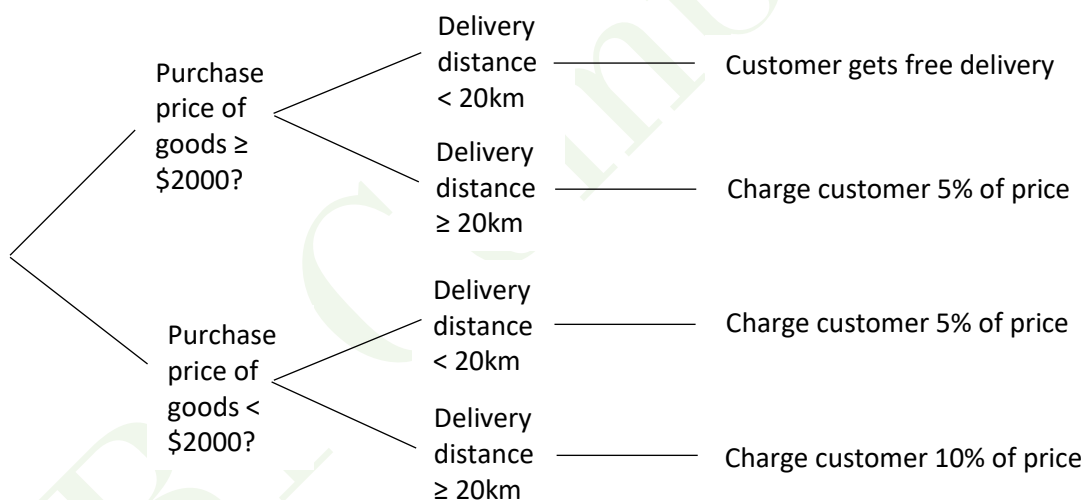
- (ii) Draw a flowchart for the new design.

6. 9754/2009/01/07 [data type, procedures, functions]

A programmer has to write a program for a large business. The program is to process and store payroll data about the employees of the business.

- (a) State three elementary data types the programmer will need to use in the program, giving an example of the use of each.
- (b) The programmer implements much of the code by using both procedures and functions. Two features to be included in the program are:
 - Calculating the total tax payable
 - Outputting a list of employees in alphabetical order
 - (i) Explain the difference between procedures and functions.
 - (ii) For each of the two features described above state whether a procedure or function would be used. Justify your answers.
- (c) Parameters can be passed to a procedure by using pass-by-value or pass-by-reference. Explain the difference between these two methods.

7. [decision table] The following decision tree shows the delivery charges made by a furniture-manufacturing company. Show the same information making use of a decision table.

**8. HCI/Prelim/9597/2014/02/04 [decision table]**

A retail company has a customer loyalty program that gives a range of discounts. For customers in the loyalty program between 5 to 9 years, they will receive a 5% discount. For 10 or more years, the customer will receive a 10% discount. Regardless of whether a customer is in the loyalty program, if the cumulative value of his purchases for this calendar year exceeds \$1000, he will receive a 15% discount.

Note: Aggregation of discounts is not allowed.

- (a) Create a decision table showing all the possible outcomes and results.
- (b) Simplify your decision table by removing redundancies.

9. 9597/2013/02/05 [decision table, pseudo-code]

Bank customers are allowed to withdraw money from their accounts at an ATM. They cannot withdraw more than the current balance in their account. There is a daily limit on the amount that can be withdrawn. In some circumstances a charge is made for the transaction. The rules are:

- the transaction is rejected if the withdrawal amount requested is greater than the current balance
 - the transaction is rejected if the withdrawal amount exceeds the daily limit
 - if the current balance before the transaction is carried out is less than 50 dollars then any successful transaction incurs a fixed charge
- (a) Create a decision table showing all the possible outcomes and results.
- (b) Simplify your decision table by removing redundancies.
- (c) Using your answer in (b) write a function using pseudocode. The function returns:
- -1 to indicate a rejection;
 - 0 for a charge-free successful transaction;
 - the charge for a chargeable successful transaction.

10. 9754/2007/01/04 [decision table]

A systems analyst has been employed to develop a computerised system for a school library which is currently operating manually.

The overdue list is processed in the following manner:

- If a book is overdue then a reminder letter would normally be sent.
 - However, if the book is more than five days overdue two further checks are made to see whether the reminder should be replaced by a warning letter:
 1. If the student has had a previous warning letter, the student will not only receive the warning letter but, in addition, a copy will be sent to the parents of the student.
 2. If the student has more than four books overdue, but no previous warning letter, the reminder letter is replaced by a warning letter.
- (a) Create a decision table showing all the possible outcomes and results.
- (b) Simplify your decision table by removing redundancies.

Section B (Assignment Questions)

1. Draw a flowchart for a number-guessing game which follows this algorithm. A secret number between 1 and 100 will be randomly generated for the player to guess. Player will be prompted to guess the number. If the player guesses the secret number correctly, a message will be output: “*You got it! You took x number of tries to guess the number*”, where x is the number of tries taken by the player. If the guess by the player is less than the secret number, output message: “*Your guess is too low. Try again.*”. If the guess is larger than the secret number, output message: “*Your guess is too high. Try again.*”. The player will then be prompted to guess again. The game will end when player guessed the secret number correctly.

Assume a random number function `RANDINT(a, b)` that generates a random integer between a and b (both ends inclusive) is available.

2. An algorithm `FindTotal` is given in pseudo-code.

```
MAIN
  DECLARE HighNum : INTEGER // HighNum is input by user
  DECLARE Total : INTEGER
  DECLARE i : INTEGER

  OUTPUT "Enter an integer: "
  INPUT HighNum
  Total  $\leftarrow$  0
  IF HighNum < 1
    THEN
      OUTPUT "Number not valid"
    ELSE
      FOR i  $\leftarrow$  1 to HighNum
        Total  $\leftarrow$  Total + i
        OUTPUT Total
      ENDFOR
    ENDIF
  ENDMAIN
```

- (a) Write down all the outputs in the correct order for the input below:
- (i) HighNum is 0,
 - (ii) HighNum is 5
- (b) Amend the pseudo-code to output the sum of square numbers from 1 to HighNum for each iteration of the for-loop.

3. **9754/2010/01/Q3fg [decision table]**

The rules that are used when deciding whether to offer insurance to customers and whether to offer discounts are as follows:

- If the customer has been refused insurance by another company and their car is over 10 years old then insurance is refused.
- If the customer has been refused insurance by another company and their car is not more than 10 years old then insurance without any discount is available.
- If the customer has not been refused insurance by another company and their car is over 10 years old then insurance without any discount is available.
- If the customer has not been refused insurance by another company and their car is less than 10 years old and they have made not more than three claims previously then insurance with a discount is available.

- (a) Create a decision table showing all the possible outcomes and results.
- (b) Simplify your decision table by removing redundancies.